Perform sentiment analysis on movie reviews using LSTM with the IMDB dataset

To develop an LSTM-based sentiment analysis model using the IMDB movie review dataset to classify reviews as positive or negative.

Algorithm

1. Import TensorFlow and the required Keras libraries.

2. Load the IMDB movie review dataset.
3. Restrict the vocabulary to the 10,000 most frequent words.
4. Pad or truncate reviews to a fixed length of 200 words.
5. Create an Embedding layer to represent words as vectors.
6. Add an LSTM layer to learn sequential information from reviews.
7. Add a sigmoid output layer for binary sentiment classification.
8. Compile the model using Adam and binary cross-entropy.
9. Train the model using the training review dataset.
10. Validate the model using a portion of the training data.
11. Evaluate the trained model using the test dataset.
12. Display the test accuracy and predict the sentiment of a sample review.

Program

```
import tensorflow as tf
from tensorflow.keras import layers, models
from tensorflow.keras.datasets import imdb
from tensorflow.keras.preprocessing.sequence import pad_sequences
tf.keras.utils.set_random_seed(28)
# Vocabulary size
vocab_size = 10000
max_len = 200
# Load IMDB dataset
(x_train, y_train), (x_test, y_test) = imdb.load_data(
num_words=vocab_size
)
# Use subsets for faster execution
x_train = x_train[:10000]
y_train = y_train[:10000]
x_test = x_test[:3000]
y_test = y_test[:3000]
# Padding
x_train = pad_sequences(
x_train,
maxlen=max_len
)
x_test = pad_sequences(
x_test,

maxlen=max_len
)
# Create LSTM model
model = models.Sequential([
layers.Embedding(vocab_size, 64),
layers.LSTM(64, dropout=0.2),
layers.Dense(1, activation='sigmoid')
])
# Compile
model.compile(
optimizer='adam',
loss='binary_crossentropy',
metrics=['accuracy']
)
# Train
model.fit(
x_train, y_train,
epochs=3,
batch_size=128,
validation_split=0.2
)
# Evaluate
loss, accuracy = model.evaluate(
```

```
x_test, y_test,
verbose=0
)
# Predict sample review
prediction = model.predict(
x_test[:1],
verbose=0
)[0][0]
print("Test Accuracy:", accuracy * 100, "%")
if prediction >= 0.5:
print("Sentiment: Positive")
else:
print("Sentiment: Negative")
```

Output of Program

```
Epoch 1/3
Epoch 2/3
Epoch 3/3
Test Accuracy: Training-dependent value
Sentiment: Positive
```

Result

Thus, an LSTM-based sentiment analysis model was successfully implemented using the IMDB movie review dataset. The model learned sequential information and classified movie reviews into positive and negative categories.